HCMS Corrected ERD

Sprint 7 reconciliation - Version 1.2 - Identity and access

users

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
full_name	VARCHAR(150)	NN
email	VARCHAR(255)	UQ, NN
password_hash	VARCHAR(255)	NN
role	ENUM	PATIENT / DOCTOR / NGO / ADMIN; NN
account_status	ENUM	ACTIVE / INACTIVE / SUSPENDED; NN
phone	VARCHAR(30)	NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

patients

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
user_id	BIGINT UNSIGNED	FK users.id, UQ, NN
national_id	VARCHAR(9)	UQ, NN
date_of_birth	DATE	NN
gender	ENUM	MALE / FEMALE; NN
address	VARCHAR(255)	NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

doctors

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
user_id	BIGINT UNSIGNED	FK users.id, UQ, NN
specialization	VARCHAR(100)	NN
license_number	VARCHAR(100)	UQ, NN
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

ngos

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
user_id	BIGINT UNSIGNED	FK users.id, UQ, NN
organization_name	VARCHAR(150)	NN
registration_number	VARCHAR(100)	UQ, NN
address	VARCHAR(255)	NULL
contact_phone	VARCHAR(30)	NN
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

join_requests

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
user_id	BIGINT UNSIGNED	FK users.id, NN
request_type	ENUM	DOCTOR / NGO; NN
status	ENUM	PENDING / APPROVED / REJECTED; NN
reviewed_by_admin_id	BIGINT UNSIGNED	FK users.id, NULL
reviewed_at	TIMESTAMP	NULL
rejection_reason	VARCHAR(500)	NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

HCMS Corrected ERD

Sprint 7 reconciliation - Version 1.2 - Clinical care

doctor_patient_assignments

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
doctor_id	BIGINT UNSIGNED	FK doctors.id, NN
patient_id	BIGINT UNSIGNED	FK patients.id, NN
status	ENUM	ACTIVE / INACTIVE; NN
assignment_reason	VARCHAR(255)	NULL
notes	TEXT	NULL
assigned_at	TIMESTAMP	NN
ended_at	TIMESTAMP	NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

medical_records

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
patient_id	BIGINT UNSIGNED	FK patients.id, UQ, NN
blood_type	VARCHAR(5)	NULL
allergies	TEXT	NULL
chronic_diseases	TEXT	NULL
medical_history	TEXT	NULL
previous_surgeries	TEXT	NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

medical_notes

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
medical_record_id	BIGINT UNSIGNED	FK medical_records.id, NN
doctor_id	BIGINT UNSIGNED	FK doctors.id, NN
note	TEXT	NN
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

treatment_plans

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
patient_id	BIGINT UNSIGNED	FK patients.id, NN
doctor_id	BIGINT UNSIGNED	FK doctors.id, NN
title	VARCHAR(150)	NN
general_instructions	TEXT	NULL
status	ENUM	ACTIVE / COMPLETED / CANCELLED; NN
start_date	DATE	NN
end_date	DATE	NULL; end_date >= start_date
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

treatment_medications

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
treatment_plan_id	BIGINT UNSIGNED	FK treatment_plans.id, NN
medication_name	VARCHAR(150)	NN
dosage	VARCHAR(100)	NN
frequency	VARCHAR(100)	NN
duration	VARCHAR(100)	NULL
instructions	TEXT	NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

treatment_appointments

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
treatment_plan_id	BIGINT UNSIGNED	FK treatment_plans.id, NN
appointment_date	DATETIME	NN
purpose	VARCHAR(255)	NN
location	VARCHAR(255)	NULL
status	ENUM	SCHEDULED / COMPLETED / CANCELLED; NN
notes	TEXT	NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

HCMS Corrected ERD

Sprint 7 reconciliation - Version 1.2 - Assistance and emergency

assistance_requests

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
patient_id	BIGINT UNSIGNED	FK patients.id, NN
doctor_id	BIGINT UNSIGNED	FK doctors.id, NN
ngo_id	BIGINT UNSIGNED	FK ngos.id, NULL
request_type	VARCHAR(100)	NN
description	TEXT	NN
status	ENUM	PENDING / UNDER_REVIEW / APPROVED / REJECTED / COMPLETED; NN
submitted_at	TIMESTAMP	NN
reviewed_at	TIMESTAMP	NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

emergency_alerts

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
patient_id	BIGINT UNSIGNED	FK patients.id, NULL
phone	VARCHAR(30)	NN
location	VARCHAR(255)	NULL
assigned_doctor_id	BIGINT UNSIGNED	FK doctors.id, NULL
assigned_by_admin_id	BIGINT UNSIGNED	FK users.id, NULL
description	TEXT	NULL
priority	ENUM	HIGH / CRITICAL; NN
status	ENUM	NEW / PENDING_ASSIGNMENT / ASSIGNED / IN_PROGRESS / RESOLVED / CANCELLED; NN
sent_at	TIMESTAMP	NN
assigned_at	TIMESTAMP	NULL
responded_at	TIMESTAMP	NULL
resolved_at	TIMESTAMP	NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

resource_allocations

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
assistance_request_id	BIGINT UNSIGNED	FK assistance_requests.id, NN
resource_name	VARCHAR(150)	NN; not an FK
quantity	INT UNSIGNED	NN; CHECK > 0
delivery_date	DATE	NULL
notes	TEXT	NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

HCMS Corrected ERD

Sprint 7 reconciliation - Version 1.2 - Communication and administration

messages

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
sender_id	BIGINT UNSIGNED	FK users.id, NN
receiver_id	BIGINT UNSIGNED	FK users.id, NN
message_text	TEXT	NN
is_read	BOOLEAN	NN; default FALSE
sent_at	TIMESTAMP	NN
read_at	TIMESTAMP	NULL
created_at	TIMESTAMP	NN

notifications

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
user_id	BIGINT UNSIGNED	FK users.id, NN
title	VARCHAR(150)	NN
message	TEXT	NN
type	VARCHAR(50)	NN
is_read	BOOLEAN	NN; default FALSE
reference_type	VARCHAR(50)	Polymorphic, NULL
reference_id	BIGINT UNSIGNED	Polymorphic, NULL
created_at	TIMESTAMP	NN
read_at	TIMESTAMP	NULL

reminders

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
user_id	BIGINT UNSIGNED	FK users.id, NN
type	ENUM	MEDICATION / APPOINTMENT; NN
reference_type	VARCHAR(50)	Polymorphic, NN
reference_id	BIGINT UNSIGNED	Polymorphic, NN
remind_at	DATETIME	NN
status	ENUM	PENDING / SENT / CANCELLED; NN
sent_at	TIMESTAMP	NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

activity_logs

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
user_id	BIGINT UNSIGNED	FK users.id, NULL
action	VARCHAR(50)	NN
entity_type	VARCHAR(50)	Polymorphic, NULL
entity_id	BIGINT UNSIGNED	Polymorphic, NULL
description	TEXT	NN
ip_address	VARCHAR(45)	NULL
created_at	TIMESTAMP	NN

system_settings

Field	Data type	Constraint / values
id	BIGINT UNSIGNED	PK, AI, NN
setting_key	VARCHAR(100)	UQ, NN
setting_value	TEXT	NULL
description	VARCHAR(255)	NULL
updated_by_admin_id	BIGINT UNSIGNED	FK users.id, NULL
created_at	TIMESTAMP	NN
updated_at	TIMESTAMP	NN

HCMS Relationships and Reconciliation

Explicit foreign-key relationships and approved Sprint 7 business rules

Approved foreign-key relationships

- patients.user_id -> users.id
- doctors.user_id -> users.id
- ngos.user_id -> users.id
- join_requests.user_id -> users.id
- join_requests.reviewed_by_admin_id -> users.id
- doctor_patient_assignments.doctor_id -> doctors.id
- doctor_patient_assignments.patient_id -> patients.id
- medical_records.patient_id -> patients.id
- medical_notes.medical_record_id -> medical_records.id
- medical_notes.doctor_id -> doctors.id
- treatment_plans.patient_id -> patients.id
- treatment_plans.doctor_id -> doctors.id
- treatment_medications.treatment_plan_id -> treatment_plans.id
- treatment_appointments.treatment_plan_id -> treatment_plans.id
- assistance_requests.patient_id -> patients.id
- assistance_requests.doctor_id -> doctors.id
- assistance_requests.ngo_id -> ngos.id
- resource_allocations.assistance_request_id -> assistance_requests.id
- emergency_alerts.patient_id -> patients.id
- emergency_alerts.assigned_doctor_id -> doctors.id
- emergency_alerts.assigned_by_admin_id -> users.id
- messages.sender_id -> users.id
- messages.receiver_id -> users.id
- notifications.user_id -> users.id
- reminders.user_id -> users.id
- activity_logs.user_id -> users.id
- system_settings.updated_by_admin_id -> users.id

Approved business rules

1. Multiple historical doctor-patient assignments may exist for the same pair, but only one ACTIVE assignment is allowed at a time.
2. Only one PENDING join request is allowed for the same user at a time.
3. Messaging is allowed only between a Patient and a Doctor who has an ACTIVE assignment with that Patient.
4. Emergency assignment is allowed only to an ACTIVE Doctor specialized in General Medicine or Emergency Medicine.
5. Doctor and NGO accounts are activated only after Admin approval; Patient accounts do not require a Join Request.
6. Notification and Reminder reference_type/reference_id values are validated in the Backend and are not direct foreign keys.
7. Activity Log entity_type/entity_id values are validated in the Backend and are not direct foreign keys.
8. Sprint 7 Emergency Alerts: patient_id is nullable for public Visitor alerts; phone is required; location and description are optional; application-created alerts start at PENDING_ASSIGNMENT.

Corrections applied to the previous ERD

1. Removed undocumented emergency_alerts fields: alert_type, latitude, longitude, and cancelled_at.
2. Emergency phone is VARCHAR(30); patient_id and description are nullable; assigned_by_admin_id and assigned_at are retained.
3. Assistance Requests use doctor_id and request_type VARCHAR(100).
4. Notifications use title VARCHAR(150) and type VARCHAR(50); no sent_at column is defined.
5. Reminders require both reference_type and reference_id; sent_at remains nullable.
6. Activity Logs use action VARCHAR(50) and description TEXT NOT NULL.
7. System Settings description is VARCHAR(255).
8. Resource Allocation resource_name is VARCHAR(150), is not an FK, and no resources table exists.

Legend: PK = Primary Key; FK = Foreign Key; UQ = Unique; NN = Not Null; NULL = Nullable; AI = Auto Increment.